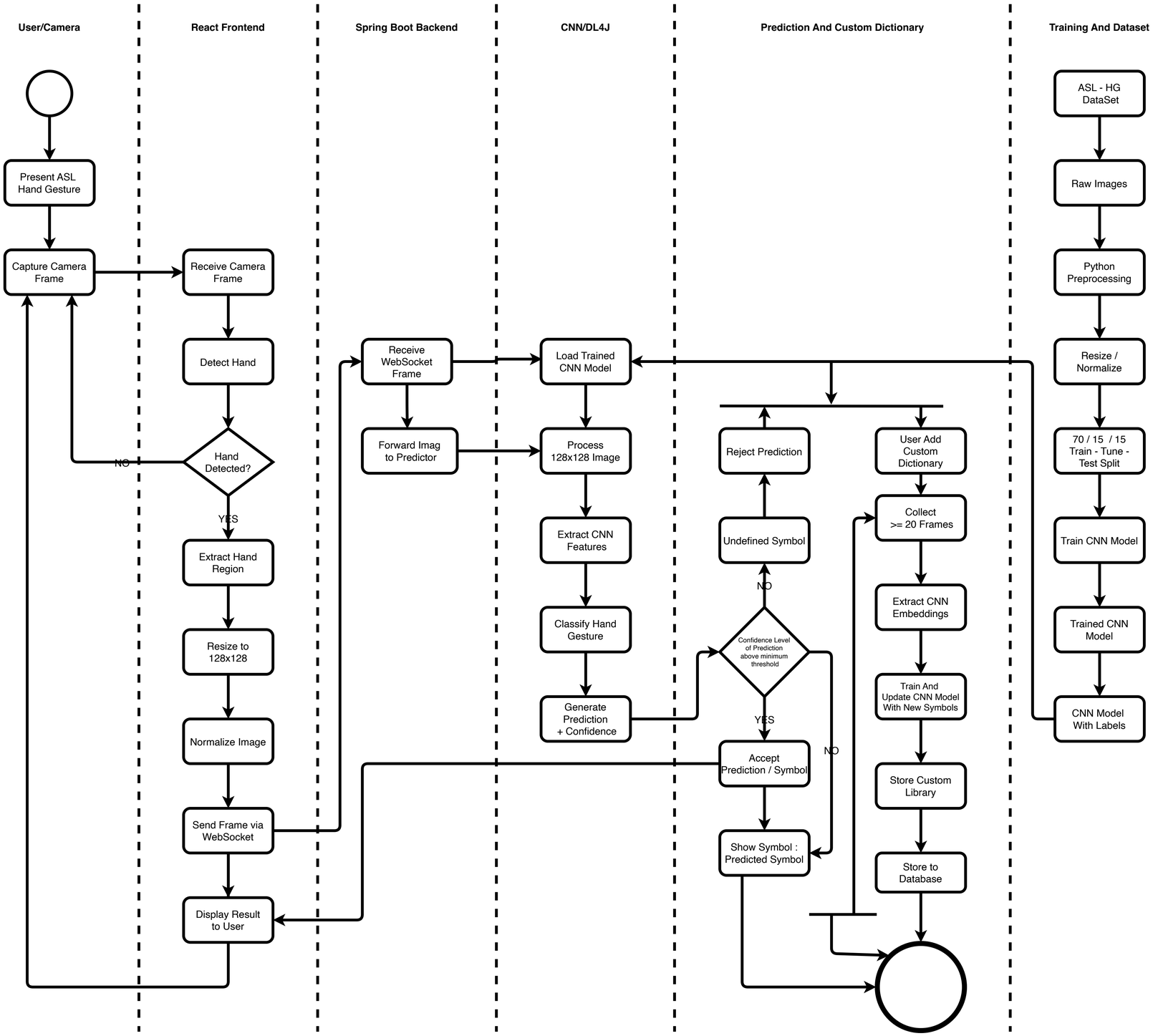
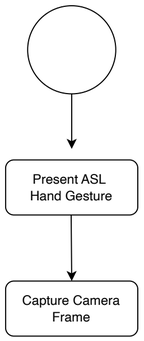
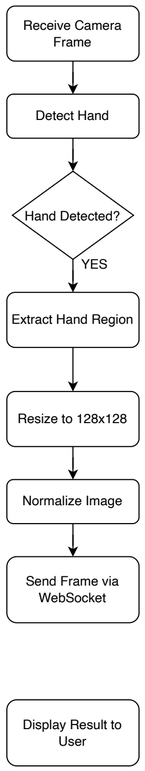

User/Camera
React Frontend
Spring Boot Backend
CNN/DL4J
Prediction And Custom Dictionary
Training And Dataset
Present ASL Hand Gesture
Capture Camera Frame
Receive Camera Frame
Detect Hand
Hand Detected?
NO
YES
Extract Hand Region
Resize to 128x128
Normalize Image
Send Frame via WebSocket
Display Result to User
Receive WebSocket Frame
Forward Imag to Predictor
Load Trained CNN Model
Process 128x128 Image
Extract CNN Features
Classify Hand Gesture
Generate Prediction + Confidence
Confidence Level of Prediction above minimum threshold
NO
YES
Accept Prediction / Symbol
Show Symbol : Predicted Symbol
Reject Prediction
Undefined Symbol
NO
User Add Custom Dictionary
Collect >= 20 Frames
Extract CNN Embeddings
Train And Update CNN Model With New Symbols
Store Custom Library
Store to Database
ASL - HG DataSet
Raw Images
Python Preprocessing
Resize / Normalize
70 / 15 / 15 Train - Tune - Test Split
Train CNN Model
Trained CNN Model
CNN Model With Labels

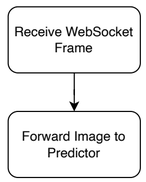
User/Camera



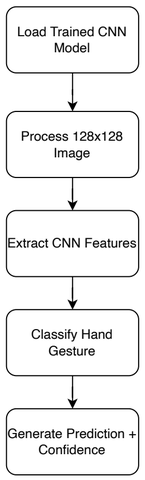
React Frontend

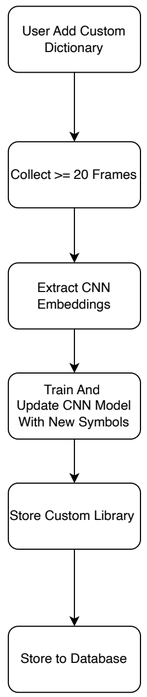


Spring Boot Backend



CNN/DL4J





Training And Dataset

